```
= V4 Outline MultiLine NoSorting TabWidth=30

H="directory"

// Global Macros use $ symbol to be called.

global nhats "D:\NHATS\Shared\base_data\NHATS cleaned"
global medi "D:\NHATS\Shared\base_data\CMS_claims\Stata"

//Intermediate Data Path
global intpath "D:\NHATS\Projects\Ankuda_RP1\data\int_data"

// Final Data Path
global datapath "D:\NHATS\Projects\Ankuda_RP1\data\final_data"

//Log files path
global outpath
"D:\HRS\Projects\surgery\zc_surgery_and_serious_illness\output\in_progres
s"

PROJECT GOAL:
Examine incident dementia via Medicare claims data vs NHATS annual
cognitive assessment

1. nhats 2011-2018
2. incident claims-based dementia cohort:
      a. ffs Medicare for 2011 & 2012
      b. no-claims-based diagnosis of dementia from 2011 to 2012
      c. 1+ claims-based diagnosis of dementia from 2013 to 2018
3. NHATS variable: prob_dem


H="get study sample"

KNOWLEDGE POINT 1: CROSSWALK NHATS TO CLAIMS DATA


use "${nhats}\sp_round_1_10.dta",clear

//exclude sample replenished in wave 5
drop if yearsample==2015

//create an index year from nhats interview date
gen index_year=ivw_year
gen index_date=ivw_date
save "${intpath}\nhats_index.dta",replace

//xwalk nhats and claims data
merge m:1 spid using "${medi}\xwalk_2018.dta"
keep if _merge==3
drop _merge
save "${intpath}\nhats_index_xwalk.dta",replace

//create ffs
```

```
use "${medi}\mbsf_06_18.dta",clear
gen ffs=bene_hmo_cvrage_tot_mons==0
save "${intpath}\mbsf_ffs.dta",replace

//merge nhats and mbsf
use "${intpath}\nhats_index_xwalk.dta",clear
joinby bid_nhats_1 using "${intpath}\mbsf_ffs.dta"
duplicates report bid_nhats_1 wave
return list
duplicates drop bid_nhats_1 wave,force

save "${intpath}\nhats_ffs.dta",replace


H="merge dx codes from claims to nhats"
```

KNOWLEDGE POINT 2: GET ICD-BASED DIAGNOSIS FROM THE CLAIMS
ip: ip_06_18
other claims data: x_09_18

```
//get icd-based dementia from the claims
foreach x in ip sn hh hs op pb dm {
      di "`x'"
      if "`x'"=="ip" {
      use bid_nhats_1 admit_date icd_dgns_cd* using
"D:/NHATS/Shared/base_data/CMS_claims/Stata/`x'_06_18.dta", clear
      }
      else {
      use bid_nhats_1 admit_date icd_dgns_cd* using
"D:/NHATS/Shared/base_data/CMS_claims/Stata/`x'_09_18.dta", clear
      }
      gen file_origin="`x'"
      rename icd_dgns_cd* dgnscd*
      rename dgnscd* diag*
      gen dementia_claims=0
      save "${intpath}\claims_`x'.dta",replace
      }
```

diagnosis 1-12: pb; dm
dignosis 1-25: all other claims data

```
foreach y in ip sn hh hs op pb dm {
      if "`y'"=="pb" | "`y'"=="dm" {
      use "${intpath}\claims_`y'.dta",clear
      foreach i of varlist diag1-diag12 {
      replace dementia_claims=1 if
inlist(`i',"F0150","F0151","F0280","F0281","F0390","G3101","G3109") |
inlist(`i',"R4181","G309","G300","G301","G308","G311","G312","G914")
      }
      }
      else {
      use "${intpath}\claims_`y'.dta",clear
      foreach k of varlist diag1-diag25 {
```

```
      replace dementia_claims=1 if
inlist(`k',"F0150","F0151","F0280","F0281","F0390","G3101","G3109") |
inlist(`k',"R4181","G309","G300","G301","G308","G311","G312","G914")
      }
      }
      keep if dementia_claims==1
      joinby bid using "${intpath}\nhats_ffs.dta"
      save "${intpath}\dx_nhats`y'.dta",replace
}
```

H="****************"

KNOWLEDGE POINT 3: CREATE CONTINUOUS FFS

```
H="indicator for ffs in 2011/2012"
*step 1: make a flag for individuals with FFS in 2011 and 2012
use "D:\NHATS\Shared\raw\CMS\NHATS CMS DUA
28016\Merged\STATA\mbsf_06_17.dta"
sort bene_id year

*make a variable that is 1/0 for 2011 based on FFS full year status
gen ffs_fullyear_2011=1 if year==2011
foreach var of varlist bene_hmo_ind_01-bene_hmo_ind_12 {
replace ffs_fullyear_2011=0 if `var'!= "0" & `var'!="4" & year==2011
}
tab ffs_fullyear_2011

*same for 2012
gen ffs_fullyear_2012=1 if year==2012
foreach var of varlist bene_hmo_ind_01-bene_hmo_ind_12 {
replace ffs_fullyear_2012=0 if `var'!= "0" & `var'!="4" & year==2012
}
tab ffs_fullyear_2012


*make a variable that is 0/1 for each person with full ffs in both 2011
and 2012
sort bene_id year
bysort bene_id: egen max_2011=max(ffs_fullyear_2011)
bysort bene_id: egen max_2012=max(ffs_fullyear_2012)
gen ffs_2011_2012=0
replace ffs_2011_2012=1 if max_2011==1 & max_2012==1
replace ffs_2011_2012=. if year!=2011
*the above step means that there is only 1 observation for each person,
and only an observation if they had an MBSF in 2011... so effectively no
duplicates per person

tab ffs_2011_2012
*now i'm going to only save this indicator
keep bene_id ffs_2011_2012
drop if ffs_2011_2012==.
*now merge to get spid and drop the bene_id
```

```
merge 1:1 bene_id using "D:\NHATS\Shared\raw\CMS\NHATS CMS DUA
28016\Crosswalks\xwalk_2016.dta", keepusing(spid)
keep if _merge==3
drop _merge

save "D:\NHATS\Projects\Ankuda_RP1\data\int_data\cka_ffs ind.dta",
replace




H="indicator for date of first dementia diagnosis"
foreach x in ip sn hh hs op pb dm {
      di "`x'"
      if "`x'"=="ip" {
      use bid_nhats_1 admit_date icd_dgns_cd* using
"D:/NHATS/Shared/base_data/CMS_claims/Stata/`x'_06_18.dta", clear
      }
      else {
      use bid_nhats_1 admit_date icd_dgns_cd* using
"D:/NHATS/Shared/base_data/CMS_claims/Stata/`x'_09_18.dta", clear
      }
      gen file_origin="`x'"
      rename icd_dgns_cd* dgnscd*
      rename dgnscd* diag*
      gen dementia_claims=0
      save "${intpath}\claims_`x'.dta",replace
      }


global intpath "D:\NHATS\Projects\Ankuda_RP1\data\int_data"
```

KNOWLEDGE POINT 4: CAPTURE ALL THE DEMENTIA DIAGNOSIS

```
foreach y in ip sn hh hs op pb dm {
      if "`y'"=="pb" | "`y'"=="dm" {
      use "${intpath}\claims_`y'.dta",clear
      foreach i of varlist diag1-diag12 {
      replace dementia_claims=1 if
inlist(`i',"3310","33111","33119","3312","3317") |
inlist(`i',"33182","33189", "2900", "29010", "29011", "29012", "29013") |
inlist(`i', "29020" "29021", "2903", "29040", "29041", "29042", "29043",
"2908") | inlist(`i', "2940", "29410", "29411", "29420", "29421", "797")
| inlist(`i',"F0150", "F0151", "F0280", "F0281", "F0390", "F0391") |
inlist(`i', "F04", "G300", "G301", "G308", "G309", "G3101", "G3109") |
inlist(`i', "G3183", "G311", "G312", "R4181")
      }
      }
      else {
      use "${intpath}\claims_`y'.dta",clear
      foreach k of varlist diag1-diag25 {
      replace dementia_claims=1 if
inlist(`k',"3310","33111","33119","3312","3317") |
inlist(`k',"33182","33189", "2900", "29010", "29011", "29012", "29013") |
```

```
inlist(`k', "29020" "29021", "2903", "29040", "29041", "29042", "29043",
"2908") | inlist(`k', "2940", "29410", "29411", "29420", "29421", "797")
| inlist(`k',"F0150", "F0151", "F0280", "F0281", "F0390", "F0391") |
inlist(`k', "F04", "G300", "G301", "G308", "G309", "G3101", "G3109") |
inlist(`k', "G3183", "G311", "G312", "R4181")
      }
      }
      keep if dementia_claims==1
      save "${intpath}\cka_dx_`y'.dta",replace
}


      *instead of save "${intpath}\dx_nhats`y'.dta",replace    save
"${intpath}\cka_dx_`y'.dta",replace

cd "${intpath}"
use cka_dx_ip.dta, clear
append using cka_dx_sn cka_dx_hh cka_dx_hs cka_dx_op cka_dx_pb cka_dx_dm

*now for each person, make a flag for the first dementia case
bysort bid_nhats_1: egen first_dem= min(admit_date)
*now only keep each observation if it is the first dementia dx
count
keep if admit_date==first_dem
count
*we don't need all the diagnosis vars, so dropping them
keep bid_nhats_1 first_dem file_origin
*we seem to have some duplicates, I'm not sure why
sort bid_nhats_1 file_origin first_dem
quietly by bid_nhats_1 file_origin first_dem: gen dup=cond(_N==1,0,_n)
keep if dup==0 | dup==1
drop dup
*there are also duplicates where there is the same data but 2 different
files-- like same date for IP and PB, for IP and OP. I'm going to just
drop the first one. from approach below I can see that out of 3,626
unique individuals, 312 have 2 clams, 5 have 3
bysort bid_nhats_1: gen n=_n
tab n
keep if n==1
drop n

*swap out bid for spid
merge 1:1 bid_nhats_1 using "D:\NHATS\Shared\raw\CMS\NHATS CMS DUA
28016\Crosswalks\xwalk_2018.dta"
keep if _merge==3
drop _merge
drop bid_nhats_1

save "${intpath}\cka_first dem.dta",replace


H="merging those claims-based indicators with nhats"
```

```
*now put it all together: merge nhats with indicator for ffs in 2011/2012
and indicator for first dementia-- going to use rounds 1-8 bc we only
have claims data through 2018

use "D:\NHATS\Shared\base_data\NHATS cleaned\sp_round_1_8.dta",clear
merge m:1 spid using "D:\NHATS\Projects\Ankuda_RP1\data\int_data\cka_ffs
ind.dta"
drop _merge
merge m:1 spid using
"D:\NHATS\Projects\Ankuda_RP1\data\int_data\cka_first dem.dta"
drop _merge



H="variable cleaning and analysis"
use "D:\NHATS\Projects\Ankuda_RP1\data\final_data\cka_working
data.dta",clear


*now work on making a "keep"indicator for people in the study- those in
2011 cohort who had ffs in 2011 and 2012. will also limit to those in
wave 1 so we can see the number of individuals, not number of
observations
gen keep=0
replace keep=1 if yearsample==2011 & ffs_2011_2012==1
replace keep=0 if wave!=1
tab keep
*we start with 5,346 individuals

*now make dementia category indicator: 1) dementia from first 2 years, 2)
incident dementia, 3) never dementia. note that there are 19,359 days
between 1/1/1960 and 1/1/2013
format first_dem %td

gen claims_dementia_cat=.
replace claims_dementia_cat=3 if first_dem==.
replace claims_dementia_cat=1 if first_dem!=. & first_dem<19359
replace claims_dementia_cat=2 if first_dem!=. & first_dem>19359
label define dc 1 "prevalent dem" 2 "incident dem" 3 "never dem"
label values claims_dementia_cat dc

tab claims_dementia_cat if keep==1

*now make variable for ever any survey-based dementia
sort spid wave
bysort spid: egen any_dem=max(prob_dem)

tab claims_dementia_cat if keep==1
tab any_dem if keep==1

*graph to show percentage of dem in nhats survey in each claims dem
category
*add tracker weight(2011 cohort)
svyset varunit [pweight=anfinwgt], strata(varstrat)
```

```
//subpop
*unweighted
tabulate any_dem claims_dementia_cat if keep==1,col
*weighted
svy,subpop(keep): tabulate any_dem claims_dementia_cat,col

//overall pop
*unweighted
tabulate any_dem claims_dementia_cat,col
*weighted
svy: tabulate any_dem claims_dementia_cat,col

graph hbar (percent)[pweight=tr2011wgt], over(any_dem, relabel(1 "no
survey dem" 2 "has survey dem")) by(claims_dementia_cat) blabel(total,
format(%9.2f) gap(*.1)) ytitle("Weighted percentage of NHATS and claims-
based dementia (full sample)")



 svy,subpop(keep): tabulate any_dem
 svy,subpop(keep): tabulate  claims_dementia_cat

H="*******************"


H="**DONT RUN**relevant code how prob_dem was created"
********DONT RUN THE TOP PART********codes are taken from NHATS_Setup_1_9
as a reference

/*
//Initialize 3 category dementia variable
//probable dementia, possible dementia, no dementia

//probable = diagnosis reported or 2+ ad8 questions (proxy) or <1.5 SD
below mean

gen dem_3_cat=-1 if ivw_type!=1 //set to na if not SP interview
replace dem_3_cat=1 if sr_dementia_ever==1 //if reported directly
replace dem_3_cat=1 if dem_3_cat==. & dem_via_proxy==1 //if proxy ivw
indicates dem
replace dem_3_cat=3 if dem_3_cat==. & dem_via_proxy==0 & speaktosp==2
//proxy ind no dem

//now apply to dementia 3 category variable
replace dem_3_cat=1 if dem_3_cat==. & inlist(speaktosp,1,-1) &
inlist(domain65,2,3)
replace dem_3_cat=2 if dem_3_cat==. & inlist(speaktosp,1,-1) &
domain65==1
replace dem_3_cat=3 if dem_3_cat==. & inlist(speaktosp,1,-1) &
domain65==0

la def dem3 1"Probable dementia" 2"Possible dementia" 3"No dementia"
la val dem_3_cat dem3
```

la var dem_3_cat "Dementia likelihood, 3 categories"
*/

*create a new variable indicate how dememtia status was found out
*dem_via_proxy comes from data from the following path:
D:\NHATS\Shared\base_data\NHATS cleaned\working\round_1_8_cleanv1
gen dem_report=.
replace dem_report=0 if sr_dementia_ever==1 //if reported directly
replace dem_report=1 if dem_3_cat==. & dem_via_proxy==1 //if proxy
indicates dem
replace dem_report=1 if dem_3_cat==. & dem_via_proxy==0 & speaktosp==2
//proxy ind no dem
label define 0 "self-report dementia" 1 "survey assessment have dementia"
label variable dem_report "how dementia was found out"

H="prob_dem, time difference & demographics of 3 groups"
use "D:\NHATS\Projects\Ankuda_RP1\data\final_data\cka_working
data.dta",clear

*now work on making a "keep"indicator for people in the study- those in
2011 cohort who had ffs in 2011 and 2012. will also limit to those in
wave 1 so we can see the number of individuals, not number of
observations
gen keep=0
replace keep=1 if yearsample==2011 & ffs_2011_2012==1
replace keep=0 if wave!=1
tab keep
*we start with 5,346 individuals

*now make dementia category indicator: 1) dementia from first 2 years, 2)
incident dementia, 3) never dementia. note that there are 19,359 days
between 1/1/1960 and 1/1/2013
format first_dem %td

gen claims_dementia_cat=.
replace claims_dementia_cat=3 if first_dem==.
replace claims_dementia_cat=1 if first_dem!=. & first_dem<19359
replace claims_dementia_cat=2 if first_dem!=. & first_dem>19359
label define dc 1 "prevalent dem" 2 "incident dem" 3 "never dem"
label values claims_dementia_cat dc

tab claims_dementia_cat if keep==1

*now make variable for ever any survey-based dementia
sort spid wave
bysort spid: egen any_dem=max(prob_dem)

tab claims_dementia_cat if keep==1
tab any_dem if keep==1

KNOWLEDGE POINT 5 track time difference between claims dementia and nhats
dementia

```
tab first_dem //date of first claims dem diag
tab any_dem //if ever had nhats dem

tab ivw_date if any_dem==1

gen time_diff=.
replace time_diff=0 if first_dem<=ivw_date
replace time_diff=1 if first_dem>ivw_date
label define time 0 "claims dem first" 1 "survey dem first"
label values time_diff time
tab time_diff if keep==1

gen actual_time_diff=.
replace actual_time_diff=abs(ivw_date-first_dem)
tab actual_time_diff

gen time_diff_cat=.
replace time_diff_cat=0 if actual_time_diff<=365
replace time_diff_cat=1 if actual_time_diff >365 & actual_time_diff <1095
replace time_diff_cat=2 if actual_time_diff>=1095
label define time_cat 0 "less than or equal to one year" 1 "from 1-3
years" 2 "greater than or equal to 3 years"
label values time_diff_cat time_cat
tab time_diff_cat if keep==1
tab time_diff_cat time_diff if keep==1

*run basic demographics for the 3 groups
*age
tabstat age if keep==1, by(claims_dementia_cat)
oneway claims_dementia_cat age if keep==1

*sex
anova female claims_dementia_cat if keep==1
tabstat female if keep==1, by(claims_dementia_cat)
tab female claims_dementia_cat if keep==1,chi2

*education
anova educ_hs_ind claims_dementia_cat if keep==1
tabstat educ_hs_ind if keep==1, by(claims_dementia_cat)

*income
tabstat imputed_inc5 if keep==1, by(claims_dementia_cat)
tabulate claims_dementia_cat if keep==1, summarize(imputed_inc5)

*self-reported health
anova srh_fp claims_dementia_cat if keep==1
tabstat srh_fp if keep==1, by(claims_dementia_cat)

//demographics for claims-based dementia
preserve
tab keep claims_dementia_cat
keep if keep==1 & claims_dementia_cat==2
*concurrent
sum age
```

```
tab female
tab race_cat
tab educ_hs_ind
sum imputed_inc5
tab srh_fp
tab medicaid
tab adl_index
tab nhres
restore
*one-year after incident claims-based dementia
gen oneyearafter=ivw_date-first_dem>0 & ivw_date-first_dem<=365
tab oneyearafter if keep==1 & claims_dementia_cat==2
```